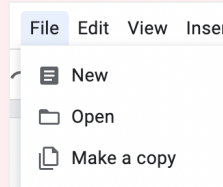


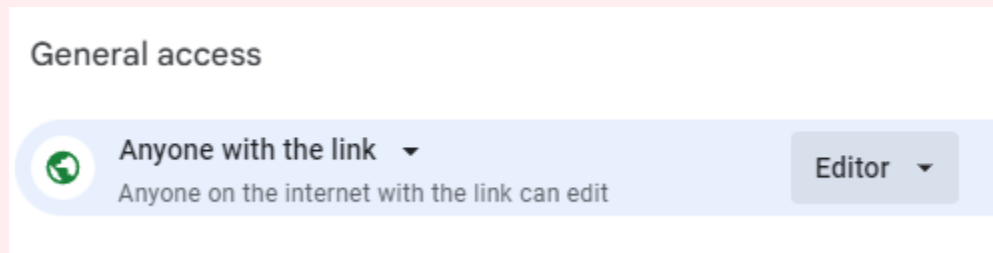
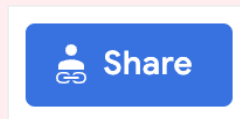
Instructions for Copying and Sharing this Document

 **DELETE THIS BOX BEFORE SUBMITTING!!**

Step 1: **Click** “File -> Make a Copy” to make a copy of this document that you can edit.



Step 2: **Change** the Share settings to “Anyone with Link -> Editor”. This will allow our graders to leave comments on your submission.



CYB102 Project 6

( [Instructions Page](#))

 Student Name: Giancarlo Montes

 Student Email: giancarlo.montes001@mymdc.net

Reflection (Required)

 **Reflection Question #1:** If I had to **explain “what is a cyber breach” in 3 emojis**, they would be...

(Feel free to put other comments about your experience in this unit here, too!)



 **Reflection Question #2:** Which step of the incident response process do you think is most important?

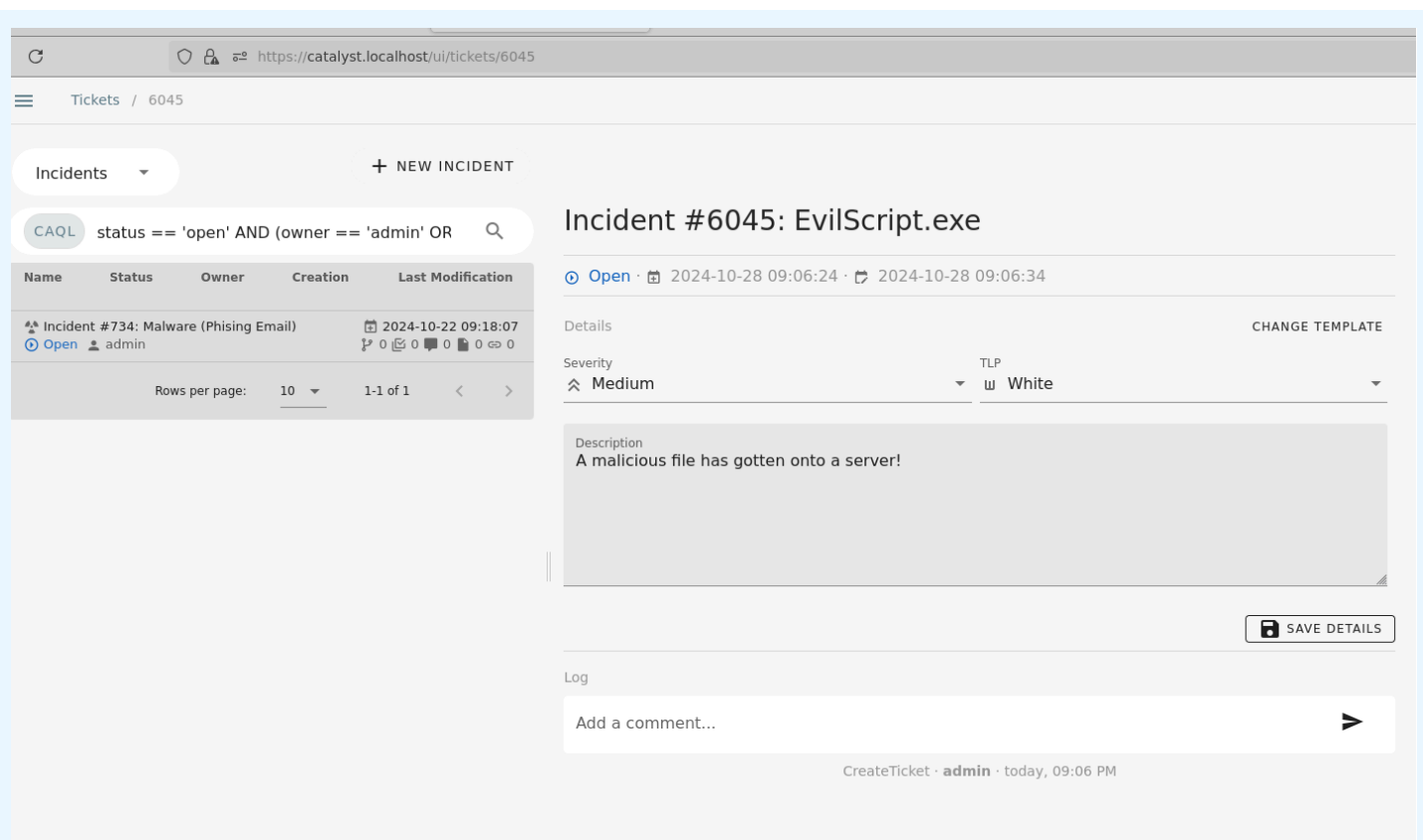
I believe the most important part of the incident response process has to be lessons learned, you can definitely learn a lot from your past mistakes and can definitely use that toward securing your system better for the future to avoid any repeats.

 **Shoutouts:** Share appreciation for anyone who helped you out with this project or made your day a little better!

Group #17

Required Challenges (Required)

Item #1: A screenshot of your Splunk Malware case in Catalyst:



The screenshot displays the Splunk Catalyst interface for incident management. The browser address bar shows the URL `https://catalyst.localhost/ui/tickets/6045`. The page header includes a menu icon, the text "Tickets / 6045", and a "+ NEW INCIDENT" button. Below the header, there is a search bar with the query `CAQL status == 'open' AND (owner == 'admin' OR` and a search icon. A table lists incidents, with the first row showing "Incident #734: Malware (Phishing Email)" and "Open" status, owned by "admin", created on "2024-10-22 09:18:07". The main content area displays "Incident #6045: EvilScript.exe" with a status of "Open" and timestamps "2024-10-28 09:06:24" and "2024-10-28 09:06:34". The "Details" section shows "Severity: Medium" and "TLP: White". The "Description" field contains the text "A malicious file has gotten onto a server!". A "Log" section at the bottom has a text input field "Add a comment..." and a "SAVE DETAILS" button. The footer shows "CreateTicket · admin · today, 09:06 PM".

New Search

1 index=pathcode EvilScript.exe Filename="EvilScript.exe"

✓ 2 events (before 10/28/24 9:00:27.000 PM) No Event Sampling ▼

Events (2) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

List ▼ Format 20 Per Page ▼

< Hide Fields

≡ All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 1

a date_month 1

a date_wday 1

date_year 1

a date_zone 1

a Event 1

a File Hash 1

a Filename 1

a Index 1

a IP 1

linecount 1

a punct 1

Response 1

a splunk_server 1

timeendpos 1

i	Time	Event
>	6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3969.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
>	6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3969.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv

6/4/23 5:59:00.000 PM 6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3969.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D

Event Actions ▼

Type	Field	Value	Actions
Selected	Event ▼	File Uploaded	▼
	File Hash ▼	3AADB7E527FC1A050E1C97FEA1CBA4D	▼
	Filename ▼	EvilScript.exe	▼
	IP ▼	192.168.1.10	▼
	Response ▼	200	▼
	host ▼	uploadedhashes	▼
	source ▼	/home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv	▼
	sourcetype ▼	csv	▼
Event	Timestamp ▼	6/4/2023 17:59	▼
	User Agent ▼	Opera/75.0.3969.218	▼
Time ⌕	_time ▼	2023-06-04T17:59:00.000-04:00	
Default	index ▼	pathcode	▼
	linecount ▼	1	▼
	punct ▼	//_..._/_..._	▼
	splunk_server ▼	lab000000	▼

Item #2: At least one artifact and notes from an external source:

1. Artifacts were:

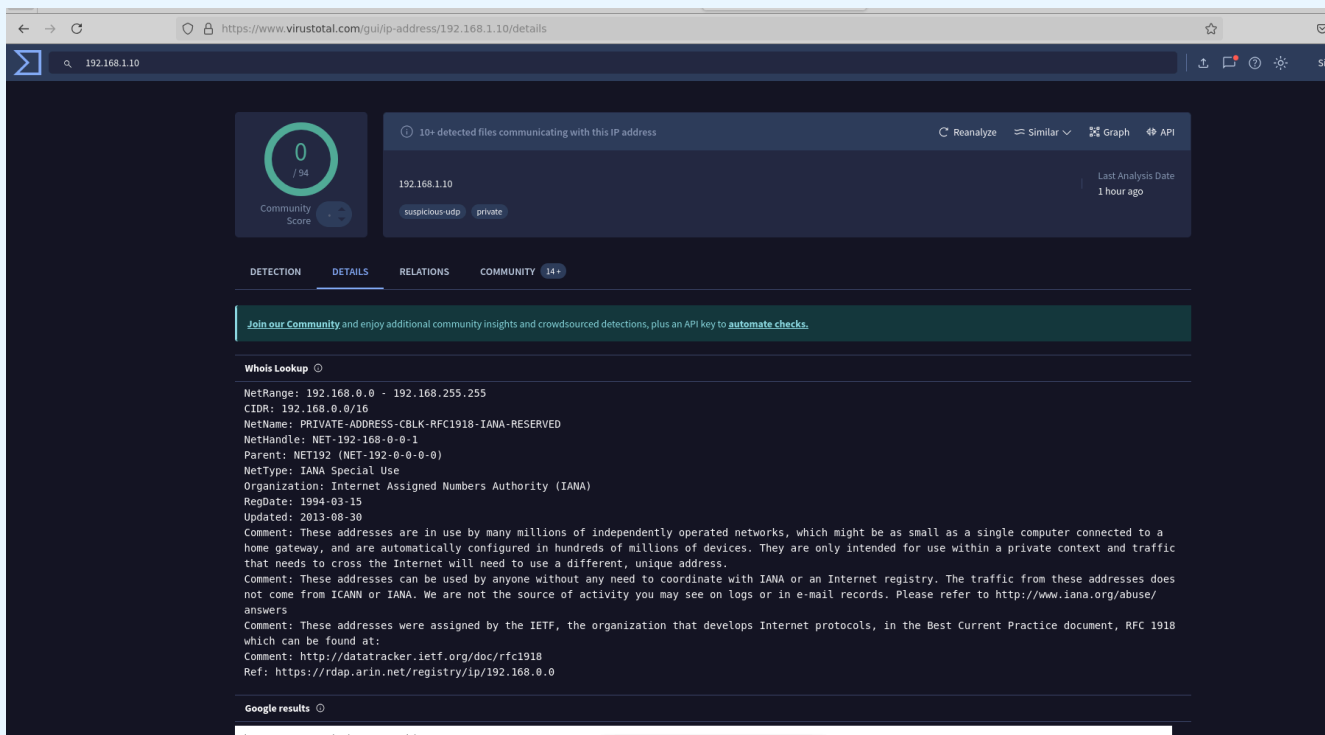
Ip: 192.168.1.10

Notes: Was reported multiple times as malicious, used as a web server attack back in

2023 according to abuseipbd, not safe!

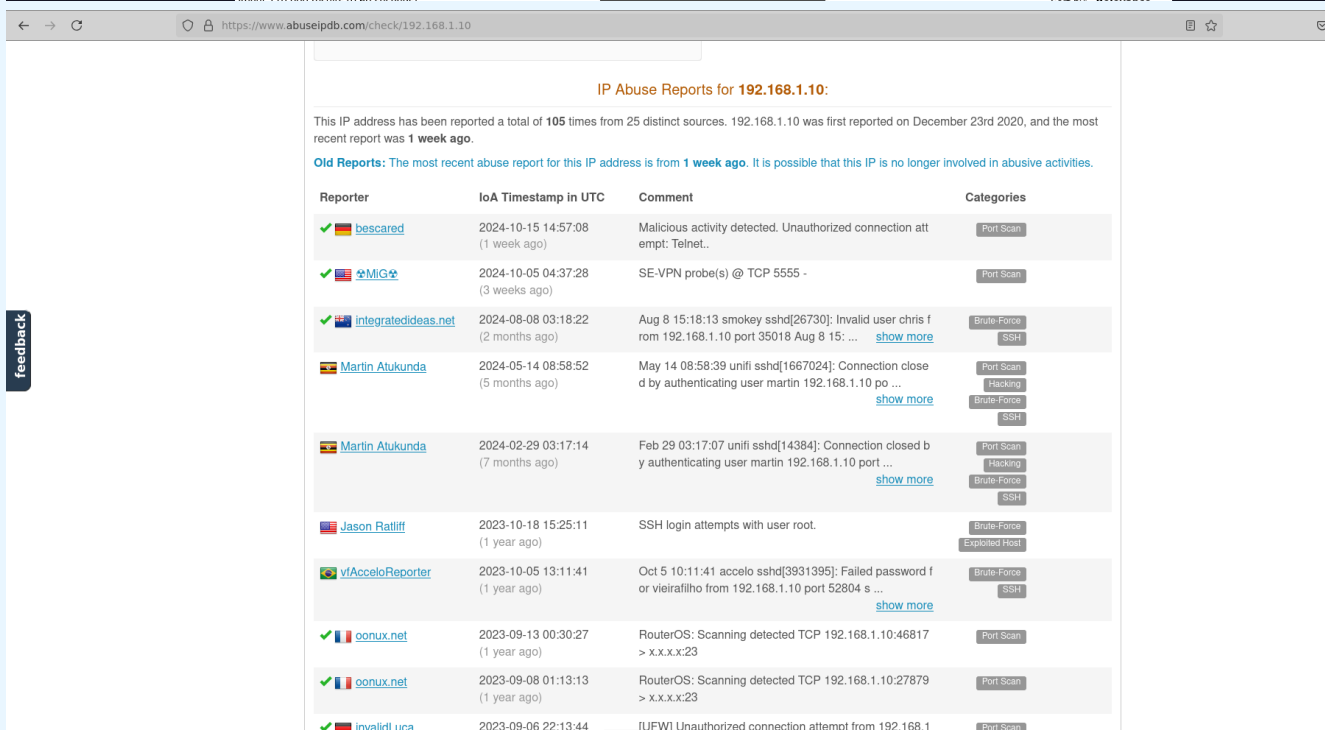
Hash: **3AADB7E527FC1A050EIC97FEA1CBA4D**

Notes: Not safe hash, reported malicious 46/63 antivirus by virustotal, and was reported to be able to unprotect documents to edit, copy a document to a specific location on the user's computer, unzip files without permission, and delete the original document.



The screenshot shows the VirusTotal interface for the IP address 192.168.1.10. At the top, there's a green circle with '0' and '/54' indicating the community score. Below this, the IP is listed as '192.168.1.10' with tags 'suspicious-udp' and 'private'. The 'Last Analysis Date' is '1 hour ago'. The 'DETECTION' tab is active, showing a 'Join our Community' message. The 'Whois Lookup' section provides details about the IP range (192.168.0.0 - 192.168.255.255), CIDR (192.168.0.0/16), and organization (IANA Special Use). It also includes a 'Google results' section with a link to a document.

2.



The screenshot shows the AbuseIPDB interface for the IP address 192.168.1.10. The page title is 'IP Abuse Reports for 192.168.1.10:'. It states that this IP address has been reported a total of 105 times from 25 distinct sources. The most recent report was 1 week ago. Below this, there's a table of 'Old Reports' with columns for Reporter, IoA Timestamp in UTC, Comment, and Categories.

Reporter	IoA Timestamp in UTC	Comment	Categories
besquared	2024-10-15 14:57:08 (1 week ago)	Malicious activity detected. Unauthorized connection attempt. Telnet...	Port Scan
MIG	2024-10-05 04:37:28 (3 weeks ago)	SE-VPN probe(s) @ TCP 5555 -	Port Scan
integratedideas.net	2024-08-08 03:18:22 (2 months ago)	Aug 8 15:18:13 smokey sshd[26730]: Invalid user chris from 192.168.1.10 port 35018 Aug 8 15: ... show more	Brute-Force SSH
Martin Atukunda	2024-05-14 08:58:52 (5 months ago)	May 14 08:58:39 unifi sshd[1667024]: Connection closed by authenticating user martin 192.168.1.10 port ... show more	Port Scan Hacking Brute-Force SSH
Martin Atukunda	2024-02-29 03:17:14 (7 months ago)	Feb 29 03:17:07 unifi sshd[14384]: Connection closed by authenticating user martin 192.168.1.10 port ... show more	Port Scan Hacking Brute-Force SSH
Jason Ratliff	2023-10-18 15:25:11 (1 year ago)	SSH login attempts with user root.	Brute-Force Exploited Host
vifAcceloReporter	2023-10-05 13:11:41 (1 year ago)	Oct 5 10:11:41 accelo sshd[3931395]: Failed password for vieirafilho from 192.168.1.10 port 52804 s ... show more	Brute-Force SSH
oonux.net	2023-09-13 00:30:27 (1 year ago)	RouterOS: Scanning detected TCP 192.168.1.10:46817 > x.x.x.x:23	Port Scan
oonux.net	2023-09-08 01:13:13 (1 year ago)	RouterOS: Scanning detected TCP 192.168.1.10:27879 > x.x.x.x:23	Port Scan
invalidLuca	2023-09-06 22:13:44	[UFW] Unauthorized connection attempt from 192.168.1	Port Scan

3.

← → ↻ https://www.virustotal.com/gui/file/208ec23c233580dbfc53aad5655845f7152ada56dd6a5c780d54e84a9d227407

208ec23c233580dbfc53aad5655845f7152ada56dd6a5c780d54e84a9d227407

Community Score

46/63 security vendors flagged this file as malicious

Reanalyze Similar More

208ec23c233580dbfc53aad5655845f7152ada56dd6a5c780d54e84a9d227407

Size: 7.95 MB Last Analysis Date: 5 days ago

docer.doc

doc create-dir auto-open detect-debug-environment calls-wmi exe-pattern handle-file run-file open-file create-ole write-file copy-file macro-run-file macros

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 9

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Code insights

The provided macros exhibit several behaviors that are commonly associated with malicious intent. Here's a detailed analysis of the macro code:

1. Obfuscated Code: The code contains several instances of obfuscation, making it difficult to understand its true purpose. For example, the `bin2var` and `var2bin` functions use poetic comments that don't provide any meaningful information about their functionality. [Show more](#)

Crowdsourced AI

ByteDefend Cyber Lab flags this file as malicious

This VBA macro code is malicious and performs a number of malicious actions, including: [Show more](#)

Hispasec flags this file as malicious

The macros extracted from the document exhibit several indicators commonly associated with malicious intent. Here is a breakdown of the behavior: [Show more](#)

Popular threat label `trojan.w97m/emoodldr` Threat categories trojan downloader dropper Family labels w97m emoodldr casdet

Security vendors' analysis

Do you want to automate checks?

4.

admin today, 09:41 PM

HASH: 3AADB7E527FC1A050E1C97FEA1CBA4D

HASH insights:

The provided macros exhibit several behaviors that are commonly associated with malicious intent. Here's a detailed analysis of the macro code:

1. Obfuscated Code: The code contains several instances of obfuscation, making it difficult to understand its true purpose. For example, the `bin2var` and `var2bin` functions use poetic comments that don't provide any meaningful information about their functionality.
2. Base64 Decoding: The `bin2var` function is used to decode a base64-encoded string and save it as a file. This technique is often used by malware to hide malicious payloads.
3. Suspicious Function Calls: The macro calls the `Shell` function to execute external commands, including copying files, deleting directories, unzipping archives, and running Python scripts. These actions could potentially be used for malicious purposes, such as downloading and executing additional malware or modifying system files.
4. Manipulation of Macro Security Settings: The macro unprotects the active document using a specific password ("securePass") and then re-protects it with the same password, allowing only read-only access. This could be an attempt to bypass macro security restrictions and gain elevated privileges.
5. Persistence Mechanism: The macro creates a scheduled task named "Auto_Open" that runs the `document_open` macro every time the document is opened. This ensures that the malicious code is executed every time the document is accessed, providing a persistent presence on the system.

5.

admin today, 09:36 PM

IP: 192.168.1.10

NetRange: 192.168.0.0 - 192.168.255.255

CIDR: 192.168.0.0/16

NetName: PRIVATE-ADDRESS-CBLK-RFC1918-IANA-RESERVED

NetHandle: NET-192-168-0-0-1

Parent: NET192 (NET-192-0-0-0-0)

NetType: IANA Special Use

Organization: Internet Assigned Numbers Authority (IANA)

RegDate: 1994-03-15

Updated: 2013-08-30

Comment: These addresses are in use by many millions of independently operated networks, which might be as small as a single computer connected to a home gateway, and are automatically configured in hundreds of millions of devices. They are only intended for use within a private context and traffic that needs to cross the Internet will need to use a different, unique address.

Comment: These addresses can be used by anyone without any need to coordinate with IANA or an Internet registry. The traffic from these addresses does not come from ICANN or IANA. We are not the source of activity you may see on logs or in e-mail records. Please refer to <http://www.iana.org/abuse/answers>

Comment: These addresses were assigned by the IETF, the organization that develops Internet protocols, in their Best Current Practice document, RFC 1918 which can be found at:

Comment: <http://datatracker.ietf.org/doc/rfc1918>

Ref: <https://rdap.arin.net/registry/ip/192.168.0.0>

This IP address has been reported a total of 105 times from 25 distinct sources. 192.168.1.10 was first reported on December 23rd 2020, and the most recent report was 1 week ago.

6.

Item #3: A brief write-up of your findings and Lessons Learned:

Findings:

The malware upload was detected when the user ABurke from IP: **192[.]68[.]1[.]0** attempted to upload the malicious **EvilScript.exe** file to a company's local network. The file was identified as a Remote Access Trojan (RAT) and confirmed via its hash:

3AADB7E527FC1A050E1C97FEA1CBA4D.

Lessons Learned:

Employees should receive training to avoid downloading and uploading files from untrusted sources also, continuous monitoring for file uploads and unusual activities is a critical aspect to take into consideration. Implementing DNS filtering solutions to block access to risky domains and malicious content including IRP to ensure proper backup and malware detection strategies are in place to prevent recurrence.

Stretch Challenge (Optional)

Bonus Task #1: Catalyst Investigation – Use Catalyst to manage the incident and fill out the case with the Artifacts, Tasks, and TTPs that you researched:

The screenshot displays the Splunk Catalyst web interface. The top navigation bar includes tabs for 'Search | Splunk 9.0.4.1', 'Catalyst', 'WannaCry IOCs and Tech', and 'VirusTotal - IP address - 1'. The main content area is titled 'Incident #13724: WannaCry Ransomware Attack'. On the left, a table lists incidents with columns for Name, Status, Owner, Creation, and Last Modification. The incident details on the right include a description of the WannaCry attack, a severity of 'High', and a TLP of 'Red'. Below the description, there is a 'Log' section with a comment from 'admin' dated today at 07:10 PM. The comment lists five tasks: 1. Initial Detection and Isolation, 2. Incident Analysis, 3. Patch Management, 4. Recovery, and 5. Communication. The right sidebar contains sections for Playbooks, References, Artifacts, Related Tickets, and Files.

Bonus Task #2: NIST or Sans Framework Analysis – Write a report that outlines the steps of either the NIST or SANS framework and how it could have prevented the breach:

NIST Cybersecurity Framework Steps

1. **Identify:** Develop an organizational understanding to manage cybersecurity risk to systems, people, assets, data, and capabilities. This includes identifying critical assets, threats, and vulnerabilities.
2. **Protect:** Implement safeguards to ensure the delivery of critical services. This involves access control, data security, and awareness training.
3. **Detect:** Implement appropriate activities to identify the occurrence of a cybersecurity event. This includes continuous monitoring and anomaly detection.
4. **Respond:** Take action regarding a detected cybersecurity incident. This involves response planning, communication, and analysis.
5. **Recover:** Maintain plans for resilience and to restore any capabilities or services impaired.

due to a cybersecurity incident.

How It Could Have Prevented the Breach

1. **Identify:** By thoroughly identifying and understanding the assets and potential threats, PathCode Inc. could have recognized the critical systems that needed extra protection and the possible attack vectors.
2. **Protect:** Implementing strong access controls and data security measures could have prevented unauthorized access and ensured that sensitive data was encrypted and protected.
3. **Detect:** Continuous monitoring and anomaly detection could have identified unusual activities or unauthorized access attempts early on, allowing for quicker intervention.
4. **Respond:** Having a well-defined incident response plan would have enabled PathCode Inc. to quickly contain and mitigate the breach, minimizing damage and reducing recovery time.
5. **Recover:** A robust recovery plan would have ensured that services were quickly restored and that any vulnerabilities were addressed to prevent future incidents.

By following the NIST Cybersecurity Framework, PathCode Inc. could have strengthened its defenses, detected the breach earlier, and responded more effectively, ultimately preventing the attacker from gaining access and uploading malware.

Submission Checklist

👉 Check off each of the features you have completed. **You will only be graded on the features you check off.**

Required Challenges

- ☒ Item #1
- ☒ Item #2
- ☒ Item #3

Stretch Challenge

- ☒ Bonus Task #1
- ☒ Bonus Task #2

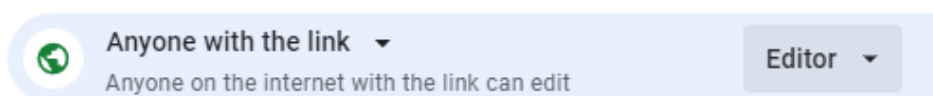
 **Tip:** You can see specific grading information, including points breakdown, by going to [the grading page](#) on the course portal.

Submit your work!

Step 1: **Click** the Share button at the top of your screen double check that anyone with the link can edit.



General access



Step 2: **Copy** the link to this document.



Step 3: **Submit** the link on the portal.